

# Matvei Borodin

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## Objective

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ML Engineer focused on LLMs, RAG, alignment, and evaluation, with strong interest in pretraining, reinforcement learning, and reliable reasoning in language models. Experienced in improving production-scale generative search systems and studying model behavior under preference- and RL-based optimization. Strong foundation in algorithms and applied ML; proficient in Python, C++, and PyTorch.

## Experience

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### Alignment Team for Alice Search Generative Answers

Moscow, Russia

ML Engineer Intern, Yandex

December 2025 — March 2026

- Worked across the **end-to-end pipeline** of generative answers for search and contributed in two directions: **development** of alignment infrastructure and **research** on RL training artifacts for the **RAG agent**.
- Developed components of the alignment and metrics infrastructure: integrated a new **factchecking reward**, implemented a **GPU whitelist** for quantized reward models, fixed versioning/build issues, corrected aggregation of retrieved information contexts produced by the **search planner**, and contributed to runtime infrastructure for **online GRPO** reward computation.
- Conducted research-oriented experiments with **CERL**, **DPO**, and **GRPO**: studied how training data composition and hyperparameters affected model behavior, analyzed **GRPO artifacts** using **side-by-side** assessor evaluations, and identified **reward hacking** patterns and quality regressions.

## Side Projects

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### A Russian Multi-hop QA Benchmark for LLM Search and Reasoning

Moscow, Russia

Pretrain YandexGPT Analytics, YSDA (School of Data Analysis)

October 2025 — Present

- Developed a Russian-language **multi-hop QA benchmark** focused on complex factual queries where LLMs must combine several constraints, reason over relations, and use search-based retrieval instead of relying only on parametric memory.
- Built a reproducible **Wikidata-based pipeline** with formal constraints, **SPARQL** queries, and gold answer sets; collected around **1,000 Russian QA items** across **20–25 domains**.
- Constructed benchmark slices targeting key LLM capabilities and failure modes: **Reversal Curse**, constraint preservation, list-style answering, hallucinations in no-answer cases, and retrieval on rare facts.
- Evaluated standalone LLMs and search-based agentic systems, including **YandexGPT Pro 5.1**, baseline models, **Alice Search**, and **NeuroAlice**, separating parametric memory, reasoning, and retrieval-based behavior.

## Education

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### Higher School of Economics

Moscow, Russia

Bachelor's Degree, Faculty of Computer Science

2022 — 2026

## Academic Achievements

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Training: Reinforcement Learning ML Run | Yandex

September 2025

Studcamp: Mathematics in AI, team optimization project | Yandex, YSDA, SPbU

November 2025

A/B Week, Yandex School of Data Analysis (YSDA) | Yandex

July 2025

Winner, All-Siberian Open School Olympiad in Mathematics

2020 — 2021

## Technical Skills

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**Languages:** Python, C++, Swift, Kotlin, Java

**Technologies:** PyTorch, SQL, Nirvana (yandex apache airflow analog), CatBoost, LoRA, Git

**Skills:** Machine Learning, Deep Learning, Classical ML, NLP, LLMs, RL (CERL, PPO, DPO, GRPO), RAG